

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE(AI)	Date:	

Code of Conduct

I T Jyothi Reddy (192472079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T Jyothi Reddy

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as “best hotels near...” and “book a flight from Chennai to...”. The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as “give the book to John,” “sleep,” and “put the file on the desk.” The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1	4	4	3	3	4	18	92
2	4	4	4	3	3	18	
3	4	4	4	3	3	18	
4	4	4	4	4	3	19	
5	3	4	4	4	4	19	

Faculty In-charge	Course Coordinator	Program Director
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Date: _____	Date: _____	Date: _____

1. CONSTITUENCY PARSING VS DEPENDENCY PARSING

Constituency parsing represents a sentence as a hierarchical tree of phrases such as noun phrases (NP) and verb phrases (VP). It shows how words are grouped into larger grammatical units.

Dependency parsing represents a sentence using direct word-to-word relationships. Each word is connected to another word, usually with the main verb as the root.

For example, in the sentence “The student solved the problem”, dependency parsing directly shows that “student” is the subject of “solved” and “problem” is its object.

Constituency parsing is useful for understanding the overall hierarchical structure, while dependency parsing is more direct for identifying grammatical relationships between words.

For conversational AI, dependency parsing is generally more suitable for extracting meaningful relationships because it directly identifies subjects, objects, modifiers, and other grammatical relationships. It is also useful for information extraction and intent understanding.

2. Top-Down Parsing vs. Earley Parsing

Top-down parsing starts from the start symbol of the grammar and tries to predict the sequence of rules that could generate the input. It works well when the input is complete and the grammar is relatively simple. However, incomplete input can cause problems because the parser may not know how much of the expected structure remains.

For example:

“best hotels near...”

The input is incomplete, but an interactive system must still determine that the user is probably beginning a location-based hotel query.

Earley parsing is a dynamic-programming parsing algorithm that can handle **CFG grammars, incomplete input, and ambiguity** more flexibly. It maintains partial parsing states and can continue parsing as new words arrive.

Aspect	Top-Down Parsing	Earley Parsing
Complete input	Works well	Works well
Incomplete input	Less flexible	Highly suitable
Ambiguity	May require additional handling	Naturally maintains alternatives
Incremental processing	Limited	Better suited
Interactive NLP	Less suitable	More suitable

For “*book a flight from Chennai to...*”, Earley parsing can maintain the partially recognized structure and wait for the destination to arrive instead of treating the unfinished query as a complete failure.

Justification: Interactive applications such as search boxes and voice assistants frequently receive **partial, incremental, or ambiguous input**. Earley parsing is therefore more appropriate because it can maintain partial hypotheses and accommodate multiple possible parses.

Conclusion: Earley parsing is the better choice for interactive NLP, particularly when users provide incomplete or ambiguous queries.

3. CFG VS PCFG VS NEURAL CONSTITUENCY PARSING

A Context-Free Grammar (CFG) represents syntactic structures using grammar rules such as:

$S \rightarrow NP VP$

CFG can represent multiple valid structures for an ambiguous sentence but does not provide probabilities for choosing between them.

A Probabilistic Context-Free Grammar (PCFG) assigns probabilities to grammar rules. When multiple parses are possible, the parser can select the most probable structure.

A neural constituency parser uses machine-learning models and contextual representations to predict syntactic structures. It can learn complex patterns from large datasets and usually provides better practical accuracy.

Comparison:

CFG:

- Simple and interpretable
- Cannot rank different interpretations

PCFG:

- Can rank possible parses
- Depends on grammar probabilities

Neural Parser:

- High practical accuracy
- Less interpretable

For a grammar-checking system, a neural constituency parser provides the best practical accuracy, while CFG or PCFG rules can be used when explicit grammatical explanations are required.

4. FEATURE STRUCTURES VS TRADITIONAL CFG

Traditional CFG represents grammatical structures using rules such as:

$S \rightarrow NP VP$

A traditional CFG mainly represents syntactic structure using rules such as:

$S \rightarrow NP VP$

However, basic CFG rules do not directly express detailed grammatical properties such as **number, gender, tense, and person**.

Feature structures extend grammar representations by attaching attributes and values to linguistic elements.

For example:

NP: {number = singular, gender = feminine, person = 3}

and

VP: {number = singular, person = 3}

Rules can then require these features to agree.

Consider:

She runs. ✓

She run. ✗

The subject *she* has **third-person singular** features. The verb must therefore have compatible agreement features.

In multilingual NLP, feature structures can represent properties such as:

- **Number:** singular/plural
- **Gender:** masculine/feminine/neuter
- **Person:** first/second/third
- **Tense:** past/present/future
- **Case:** nominative/accusative/etc.
- **Agreement:** subject–verb or noun–adjective agreement

Aspect	Traditional CFG	Feature Structures
Phrase structure	✓	✓
Number agreement	Limited	✓
Gender agreement	Limited	✓
Person agreement	Limited	✓
Tense information	Limited	✓
Multilingual constraints	Less flexible	Highly flexible

Justification: Languages differ significantly in agreement and grammatical marking. Feature structures allow the same general grammar framework to represent these language-specific constraints without creating an excessive number of separate grammar rules.

Conclusion: Feature structures are more powerful than basic CFG representations because they combine **syntactic structure with grammatical attributes**, making them highly useful for multilingual NLP.

5. SUBCATEGORIZATION FRAMES IN VOICE ASSISTANT COMMANDS

Subcategorization describes the types and number of arguments that a verb requires or permits.

Different verbs impose different syntactic requirements.

For example:

“Give”

give + object + recipient

“Give the book to John.”

Here, *give* requires:

- an object → *the book*
- a recipient → *to John*

“Sleep”

sleep + no obligatory object

“John sleeps.”

The verb *sleep* does not require a direct object.

“Put”

put + object + location

“Put the file on the desk.”

The verb *put* requires:

- an object → *the file*
- a location → *on the desk*

A simplified representation is:

Verb Subcategorization Frame Example

give V + NP + PP Give the book to John

sleep V John sleeps

put V + NP + PP Put the file on the desk

This information allows an NLP system to determine whether a sentence has the **expected argument structure**.

For example:

“Put the file.”

is syntactically incomplete in the intended sense because *put* normally requires a location.

Similarly:

“Sleep the book.”

violates the normal argument requirements of *sleep*.

Subcategorization is also important for **semantic analysis** because it helps determine the roles played by different arguments. In “*Give the book to John*,” the system can identify:

- **Agent:** person giving
- **Theme:** book
- **Recipient:** John

Justification: A voice assistant must understand not only whether a sentence is grammatically possible but also **what entities participate in the action and what roles they perform**. Subcategorization provides this bridge between syntactic structure and semantic interpretation.

Conclusion: Subcategorization frames are essential for identifying valid verb structures, detecting incomplete or incorrect commands, and extracting semantic roles. Therefore, they are highly important for both **syntactic parsing and semantic understanding** in voice assistants.